MILITARY SPECIFICATION

BALLISTICS COMPUTER SUBASSEMBLY: 8635275
(OUTPUT UNIT)

This amendment forms a part of Military Specification MIL-B-60121(MU) dated 28 May 1964.

Paragraph 2.1, Specifications: Delete "MIL-I-45208, Inspection System Requirements".

Paragraph 2.1, Drawings: Add "F8565556 - Pressure Tester - Two Station".

Paragraph 3.4.3: Delete and substitute:

"3.4.3 Sealing.- The output unit shall exhibit an internal pressure drop of less than 0.2 psig when pressurized to 5 ± 0.1 psig for a minimum of 2 hours".

Add the following paragraph:

"3.4.4 Flushing.- The output unit shall be flushed until the interior atmosphere contains nitrogen with a dewpoint at least as low as -25°F. The unit shall be sealed with an internal pressure of nitrogen of 3.0 ± .5 psig".

Add the following paragraph:

"3.4.5 Storage temperature.- The unit shall show no evidence of physical failure and shall meet all requirements of this specification at standard ambient temperature of +60° to +90°F after having been exposed to and thermally stabilized at ambient temperature of -80°F and + 160°F".

Paragraph 3.5.2: Delete "plus twice the rated voltage".

Paragraph 3.6.8: Delete " .010 volts" and substitute " .010 volts rms".

Paragraph 3.6.9.2: Add the following sentence:

"The output unit shall slew from stop to stop in both directions within a maximum running time of 9 seconds for each direction, without load on the output shaft, at -40° and +150°F".

Add new paragraph:

Paragraph 3.6.10: Add the following sentence:

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"The motor clutch shall slip when the load is increased above 9 inch pounds".

Paragraph 3.6.11: Delete "3.5 volts ac" and substitute "3.5 volts rms".

Paragraph 4.1.1: Delete and substitute:

"4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109".

Add new paragraph:

"4.1.2 First article (initial production) inspection.- When required by the contract, an initial production sample shall be submitted to a Government approved facility in accordance with the contract schedule for test and evaluation. The sample shall consist of three output units plus three of every component and subassembly covered by Supplementary Quality Assurance Provisions (SQAP). The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures, as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials. After inspection by the contractor and provisional acceptance at source by the Government, the sample shall be inspected at a Government facility or such other facility as specified in the contract, for all the requirements of the drawings and specifications".

Paragraph 4.2.2, Table II: Add the following characteristics:

"116.	Sealing	3.4.3	4.5.1.3
117.	Flushing	3.4.4	4.5.1.4".

Paragraph 4.3.2: Delete second sentence and substitute:

"Each sample shall meet the following requirements and tests and shall then meet the requirements and tests in Table II"; Delete characteristic "305"; Add the following characteristic:

"308.	Storage temperature	3.4.5	4.5.1.5".

Paragraph 4.4, first sentence: Delete "Measuring and Test Equipment Requirement of MIL-I-45208 and all".

Paragraph 4.4.1, Item Number 2 description: Delete "1 megohm ± 10%" and substitue "1 megohm minimum"; Item Number 3 description: Delete and substitute "0-130 volts rms, 400 cps ± 1%, 2% maximum harmonic distortion, 3 amperes output"; Item Number 4 description: Delete "1.0 volt peak to peak" and substitute "0.1 volt peak to peak".

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Paragraph 4.5.1.3: Delete and substitute:

"4.5.1.3 Sealing.- The special testing equipment to test sealing of the output unit shall conform to the internal pressure test requirements of Sheet 1 of Drawing F8565556, Pressure Tester-Two Station. Secure the poppet in Valve 10516717 in a closed position. The pressure shall be applied to the output unit through the input valve in the side of the unit. Pressurize the output unit with dry nitrogen gas to the value specified in 3.4.3, in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. The dry nitrogen gas shall have a dew-point at least as low as minus 25°F. Retain the applied pressure in the output unit for a minimum of 2 hours. Within the first hour of this 2 hour time period, the output shaft (ballistics drive) shall be rotated one half revolution clockwise then counterclockwise a minimum of 3 times. Any evidence of leakage in the output unit, at the end of the 2 hour time period, shall not exceed the tolerance specified in 3.4.3, as measured on the pressure gauge of the testing equipment. Upon completion of this test, release the poppet in Valve 10516717".

Add the following paragraph:

"4.5.1.4 Flushing.- Perform this test in sequence to the test in 4.5.1.3. The special testing equipment specified in 4.5.1.3 shall be utilized in this test. A dew-point tester, with an appropriate adapter to accommodate the Valve 10516717, shall be used to test the dew-point of the interior atmosphere of the output unit. The output unit shall be flushed with dry nitrogen gas. The flushing shall continue until the nitrogen gas bled from the output unit has a dew-point at least as low as minus 25°F, as measured with the dew-point tester. The dew-point tester shall be removed from the output unit. The internal pressure of nitrogen gas in the output unit shall be brought to a measured 3 psig ± 0.5 psig. The internal pressure of the output unit shall be monitored for a period of 5 minutes ± 15 seconds. At the end of the 5 minute period there shall be no evidence of leakage in the output unit as indicated on the pressure gauge of the testing equipment. The output unit shall conform to the requirements of 3.4.4.

Add the following paragraph:

"4.5.1.5 Storage temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.5. The output unit shall be exposed to each temperature extreme specified in 3.4.5 for a minimum of 5 hours. After each temperature exposure the output unit shall be stabilized in an ambient temperature between 60° and 90°F and subjected to a visual, tactile and audible examination for evidence of physical failure. The output unit shall then be subjected to the tests in 4.5.1.1 through 4.5.13 inclusive to determine conformance to 3.4.5. Caution shall be exercised during the temperature cycling to avoid subjecting the output unit to thermal shock".

Paragraph 4.5.9, 3rd sentence: After ".010 volt" add "rms".

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Paragraph 4.5.10.2: Add the following sentence:

"When this test is performed at -40° and +150°F the slewing speed of the output shaft shall not exceed 9 seconds for each direction of travel".

Paragraph 4.5.12: Delete 2nd sentence and substitute:

"This voltage shall be a minimum of 3.5 volts rms and in-phase with the voltage applied at pin L within 30°".

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